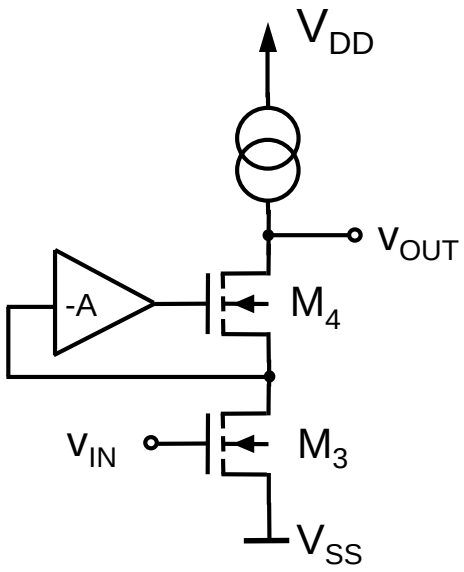
Exercise 4-3
Single-Stage Amplifiers

Winter Term 2025/26

Nima Shahpari, Ph.D.

Exercise 4-3 – Task 1

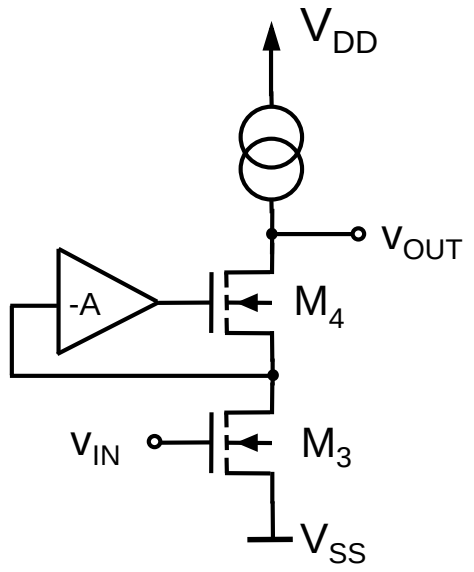
Task 1: Regulated Cascode Amplifier



Exercise 4-3 – Task 1

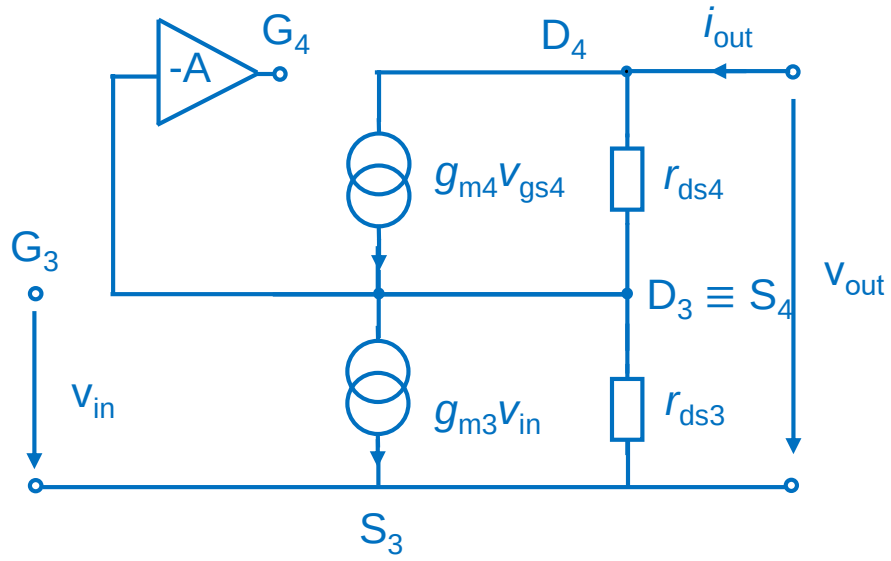
1.1. In which region are M3 and M4 operated? Why?

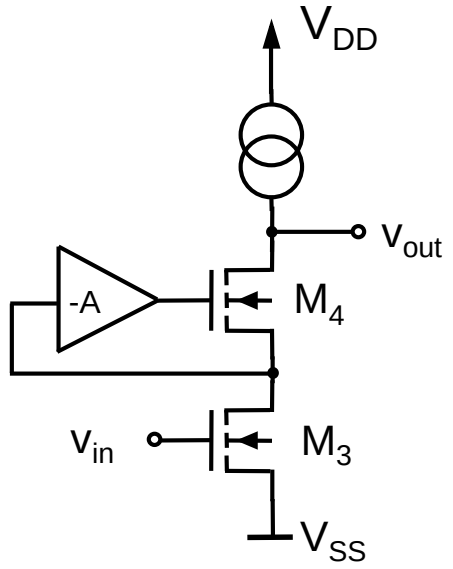
The transistors are biased in the saturation region, where r_{out} is large and $g_m \sim V_{GS}$ → large small-signal gain



Exercise 4-3 – Task 1

1.2. Draw the small signal equivalent circuit! Note that the amplifier block “-A” is to be used “as it is” in the small-signal equivalent circuit. Label all voltages, currents and transistor contacts.

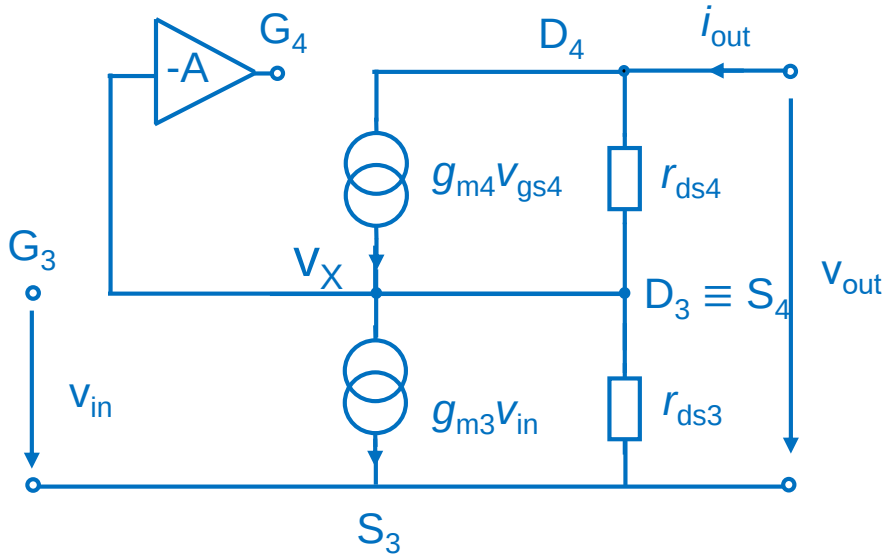


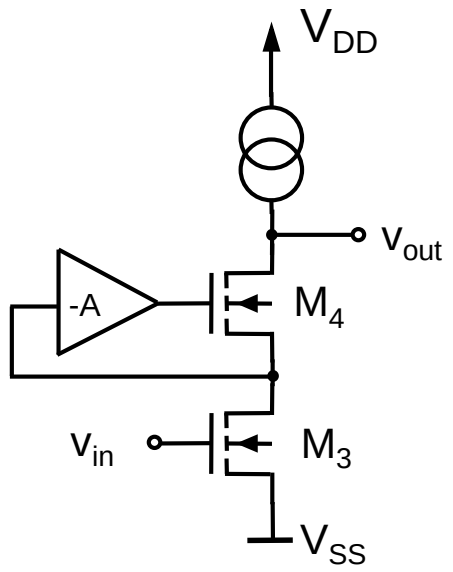


Exercise 4-3 – Task 1

1.3. Determine the transconductance $G_\mathrm{m} = \left.\frac{i_\mathrm{out}}{v_\mathrm{in}}\right|_{v_\mathrm{out}=0\,\mathrm{V}}$

$$G_\mathrm{m} = \left.\frac{i_\mathrm{out}}{v_\mathrm{in}}\right|_{v_\mathrm{out}=0\,\mathrm{V}}$$





$$v_\mathrm{gs4} = -Av_\mathrm{x} - v_\mathrm{x} = -(1+A)v_\mathrm{x}$$

$$\begin{aligned} i_\mathrm{out} &= g_\mathrm{m3}v_\mathrm{in} + \frac{1}{r_\mathrm{ds3}}v_\mathrm{x} \\ &= -\frac{1}{r_\mathrm{ds4}}v_\mathrm{x} - (1+A)g_\mathrm{m4}v_\mathrm{x} \end{aligned}$$

replace

$$v_\mathrm{x} = -\frac{g_\mathrm{m3}}{\frac{1}{r_\mathrm{ds3}} + \frac{1}{r_\mathrm{ds4}} + (1+A)g_\mathrm{m4}}v_\mathrm{in}$$

Exercise 4-3 – Task 1

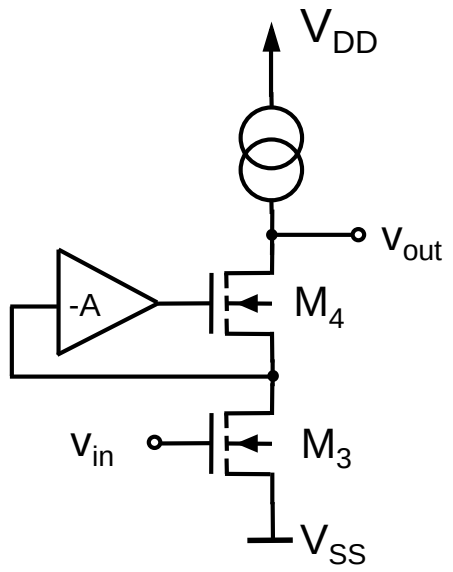
1.3. Determine the transconductance $G_\mathrm{m} = \left.\frac{i_\mathrm{out}}{v_\mathrm{in}}\right|_{v_\mathrm{out}=0\,\mathrm{V}}$

$$i_\mathrm{out} = g_\mathrm{m3} v_\mathrm{in} - \frac{1}{r_\mathrm{ds3}} \frac{g_\mathrm{m3}}{\frac{1}{r_\mathrm{ds3}} + \frac{1}{r_\mathrm{ds4}} + (1+A) g_\mathrm{m4}} v_\mathrm{in}$$

$$G_\mathrm{m} = \frac{i_\mathrm{out}}{v_\mathrm{in}} = g_\mathrm{m3} \left(1 - \frac{1}{1 + \underbrace{\frac{r_\mathrm{ds3}}{r_\mathrm{ds4}}}_{\approx 1} + \underbrace{(1+A)}_{\gg 1} g_\mathrm{m4} r_\mathrm{ds4}} \right)$$

$\ll 1$

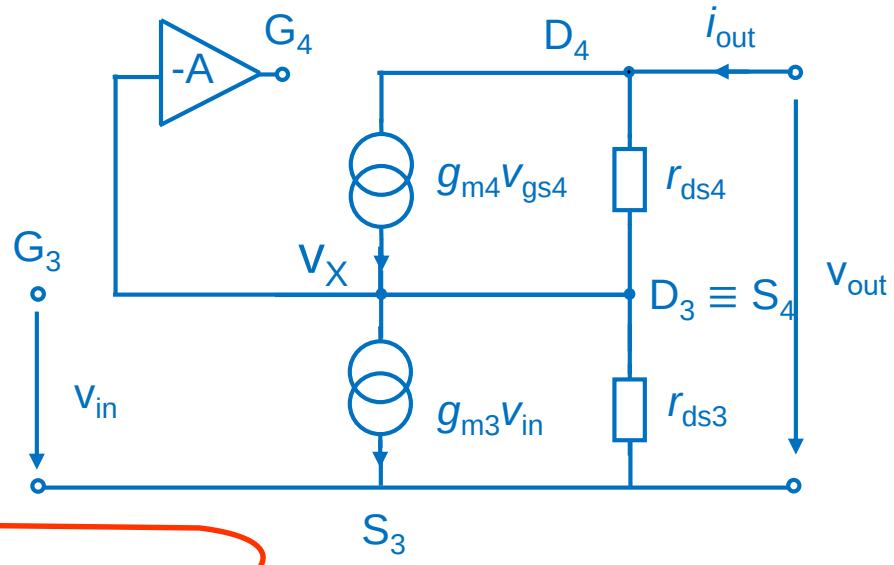
$$\boxed{G_\mathrm{m} = \frac{i_\mathrm{out}}{v_\mathrm{in}} \approx g_\mathrm{m3}}$$



Exercise 4-3 – Task 1

1.4. Determine the small-signal output resistance r_{out} of the circuit!

$$r_{\text{out}} = \left.\frac{v_{\text{out}}}{i_{\text{out}}}\right|_{v_{\text{in}} = 0\,\text{V}}$$



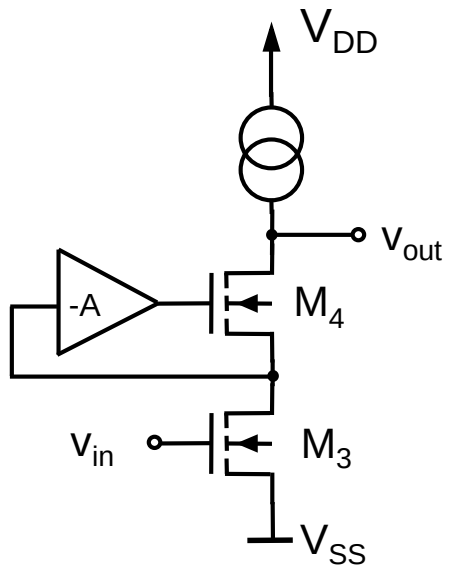
$$v_{\text{x}} = r_{\text{ds3}} i_{\text{out}}$$

replace

$$v_{\text{gs4}} = -A v_{\text{x}} - v_{\text{x}} = -(1 + A) r_{\text{ds3}} i_{\text{out}}$$

replace

$$i_{\text{out}} - g_{m4} v_{\text{gs4}} - \frac{v_{\text{out}} - v_{\text{x}}}{r_{\text{ds4}}} = 0$$



Exercise 4-3 – Task 1

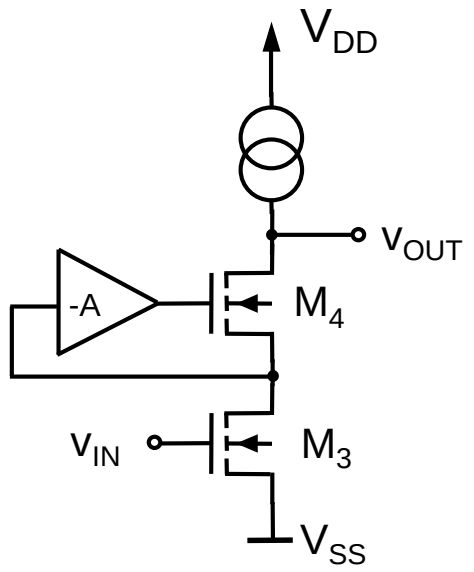
1.4. Determine the small-signal output resistance r_{out} of the circuit!

$$i_{\text{out}} + g_{\text{m4}}(1+A)r_{\text{ds3}}i_{\text{out}} - \frac{v_{\text{out}}}{r_{\text{ds4}}} + \frac{r_{\text{ds3}}}{r_{\text{ds4}}}i_{\text{out}} = 0$$

$$\frac{v_{\text{out}}}{r_{\text{ds4}}} = i_{\text{out}} + \frac{r_{\text{ds3}}}{r_{\text{ds4}}}i_{\text{out}} + g_{\text{m4}}(1+A)r_{\text{ds3}}i_{\text{out}}$$

$$\frac{v_{\text{out}}}{i_{\text{out}}} \approx 2r_{\text{ds3}} + Ag_{\text{m4}}r_{\text{ds4}}r_{\text{ds3}}$$

$$\approx Ag_{\text{m4}}r_{\text{ds4}}r_{\text{ds3}}$$



Output resistance of cascode, i.e., $g_{\text{m4}}r_{\text{ds4}}r_{\text{ds3}}$, boosted by A!

Exercise 4 – Task 1

1.5. Determine the small-signal gain $A_V = -G_m r_{out}$!

With the results of task 1.3 and task 1.4, the small-signal gain A_V results in

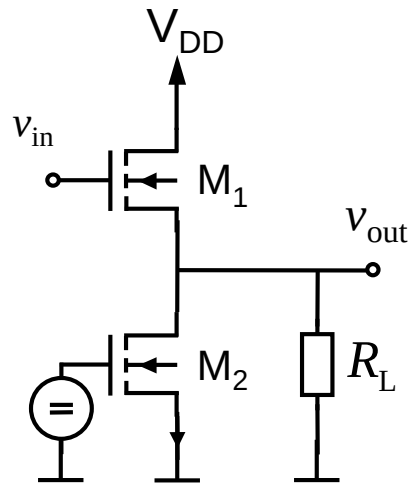
$$A_V = -G_m r_{\mathrm{out}} \approx -g_{\mathrm{m3}} A g_{\mathrm{m4}} r_{\mathrm{ds4}} r_{\mathrm{ds3}}$$

Small-signal gain of the cascode, i.e., $g_{m3} g_{m4} r_{ds4} r_{ds3}$, boosted by A!

Exercise 4-3 – Task 2

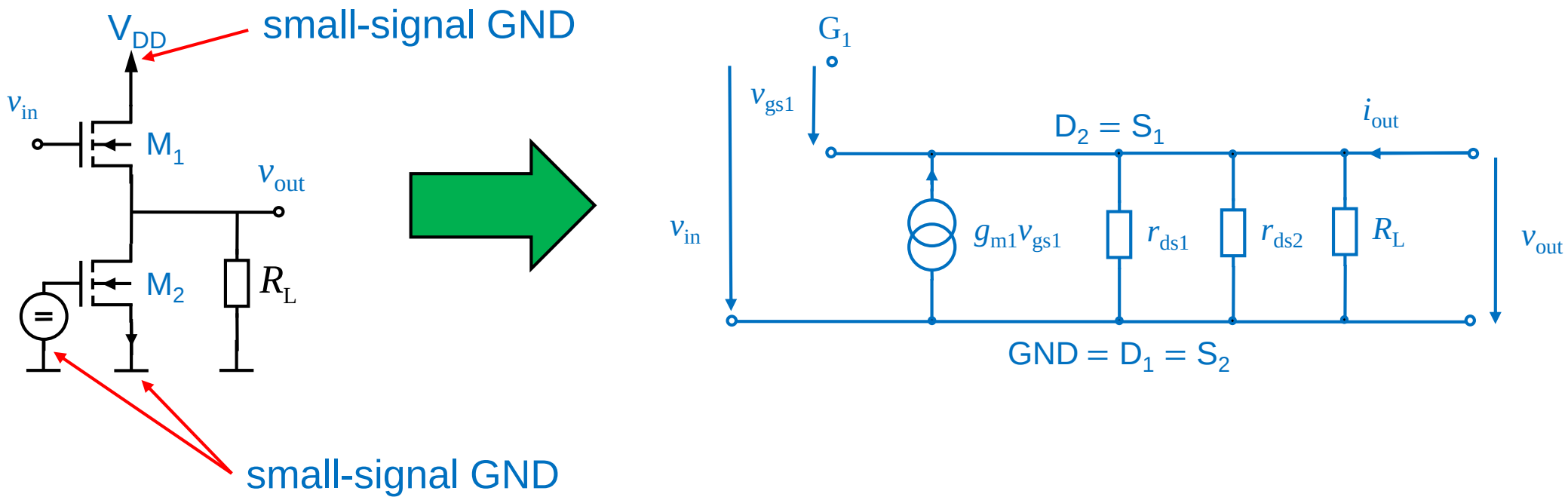
2.1. Which circuit is shown in figure 2?

source-follower
common-drain amplifier
level-shifter



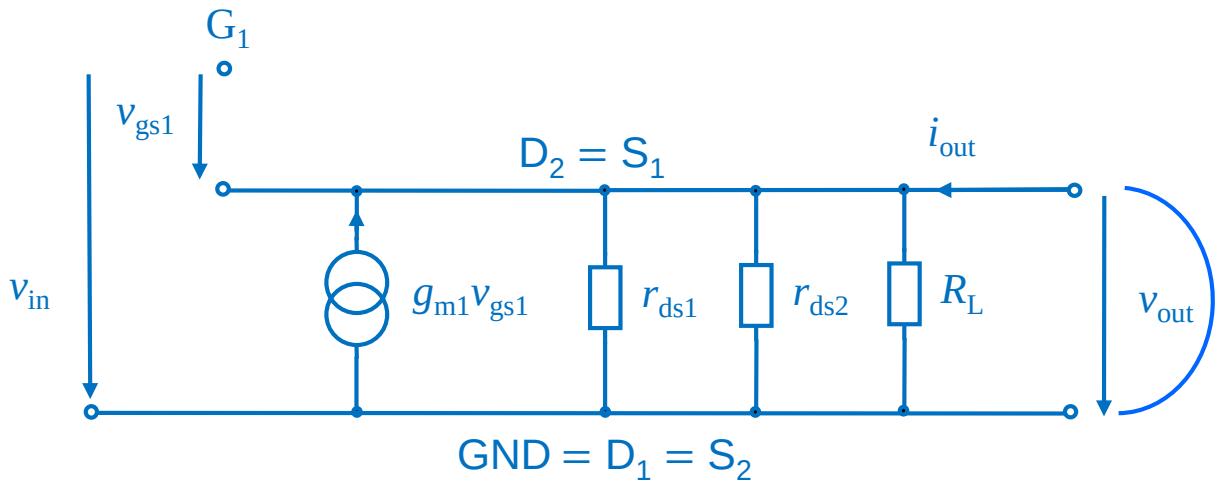
Exercise 4-3 – Task 2

2.2. Draw the small-signal equivalent circuit including the load resistance R_L! The body-effect is to be neglected. Label all voltages, currents and contacts!



Exercise 4-3 – Task 2

2.3. Determine the transconductance $G_\mathrm{m} = \left.\frac{i_\mathrm{out}}{v_\mathrm{in}}\right|_{v_\mathrm{out}=0\ \mathrm{V}}$



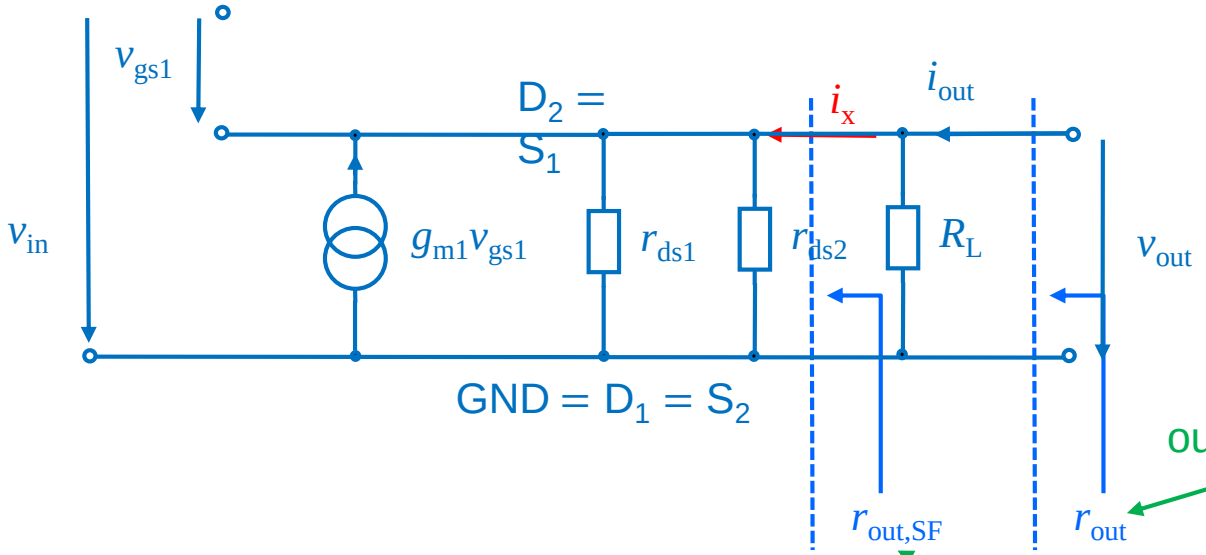
$$v_\mathrm{gs1} = v_\mathrm{in}$$

$$i_\mathrm{out} = -g_\mathrm{m1} v_\mathrm{in}$$

$$G_m = \left.\frac{i_\mathrm{out}}{v_\mathrm{in}}\right|_{v_\mathrm{out}=0\ \mathrm{V}} = -g_\mathrm{m1}$$

Exercise 4 – Task 2

2.4. Determine the small-signal output resistance r_{out} of the circuit!



$$r_{out,SF} = \left.\frac{v_{out}}{i_x}\right|_{v_{in}=0\,V} \Rightarrow \begin{cases} i_x = -g_{m1}v_{gs1} + \dfrac{v_{out}}{r_{ds1}||r_{ds2}} \\ v_{gs1} = -v_{out} \end{cases}$$

$$r_{out} = r_{out,SF}||R_L$$

$$r_{out,SF} = r_{ds1}||r_{ds2}||\frac{1}{g_{m1}} \approx \frac{1}{g_{m1}}$$

$$r_{out} = R_L||r_{ds1}||r_{ds2}||\frac{1}{g_{m1}} \approx R_L||\frac{1}{g_{m1}}$$

2.5. Determine the small-signal gain $A_V = -G_m r_{out}$!

$$G_m = -g_{m1}$$
$$r_{out} \approx R_L \| \frac{1}{g_{m1}} = \frac{R_L}{g_{m1}R_L+1}$$

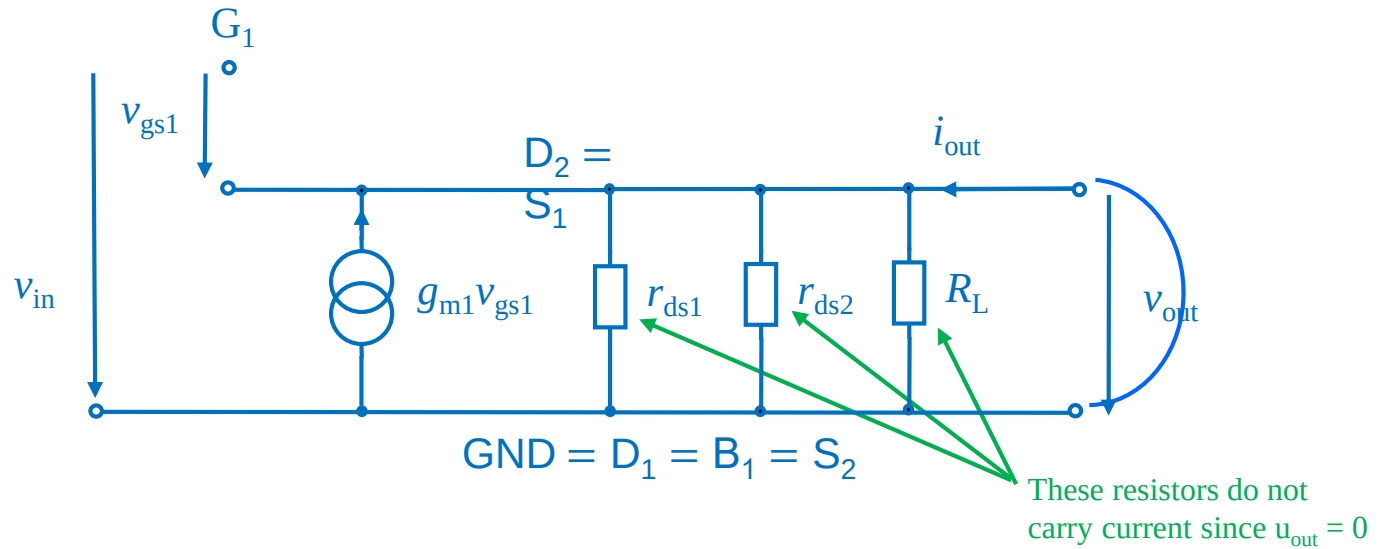
$$A_V \approx \frac{g_{m1}R_L}{g_{m1}R_L+1}$$

The maximum value of the voltage gain results in

$$A_{V_{max}} \approx \lim_{g_{m1}R_L \to \infty} \left(\frac{g_{m1}R_L}{g_{m1}R_L + 1} \right) = 1$$

Exercise 4 – Task 2

2.6. Repeat task 2.5 while considering the body-effect! Does the body-effect increase or decrease the small-signal gain of the circuit?

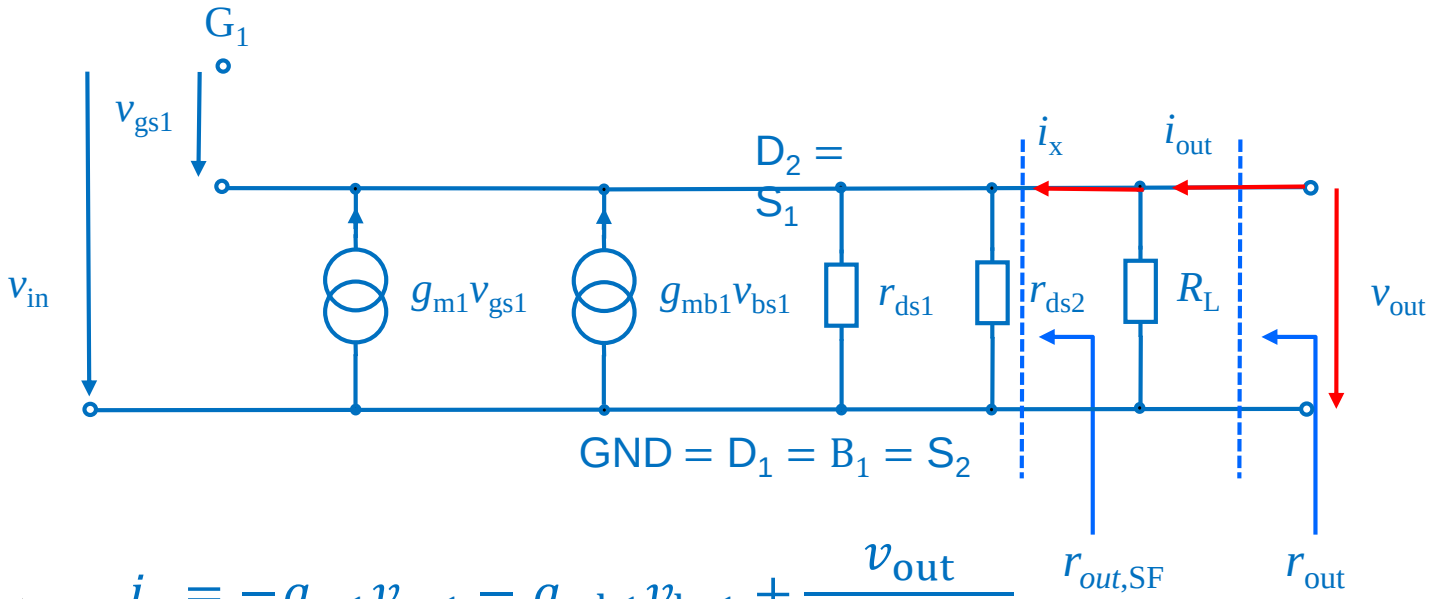


$$v_{\text{out}} = 0$$
$$v_{\text{bs1}} = -v_{\text{out}}$$
$$v_{\text{gs1}} = -v_{\text{in}}$$
$$i_{\text{out}} = -g_{\text{m1}} v_{\text{in}}$$

$$G_{\text{m}} = \left.\frac{i_{\text{out}}}{v_{\text{in}}}\right|_{v_{\text{out}} = 0\ \text{V}} = -g_{\text{m1}}$$

Exercise 4 – Task 2

2.6. Repeat task 2.5 while considering the body-effect! Does the body-effect increase or decrease the small-signal gain of the circuit?



$$r_{\text{out,SF}} = \left.\frac{v_{\text{out}}}{i_{\text{x}}}\right|_{v_{\text{in}} = 0\ \text{V}}$$

$$i_{\text{x}} = -g_{m1}v_{\text{gs1}} - g_{\text{mb1}}v_{\text{bs1}} + \frac{v_{\text{out}}}{r_{\text{ds1}} \| r_{\text{ds2}}}$$

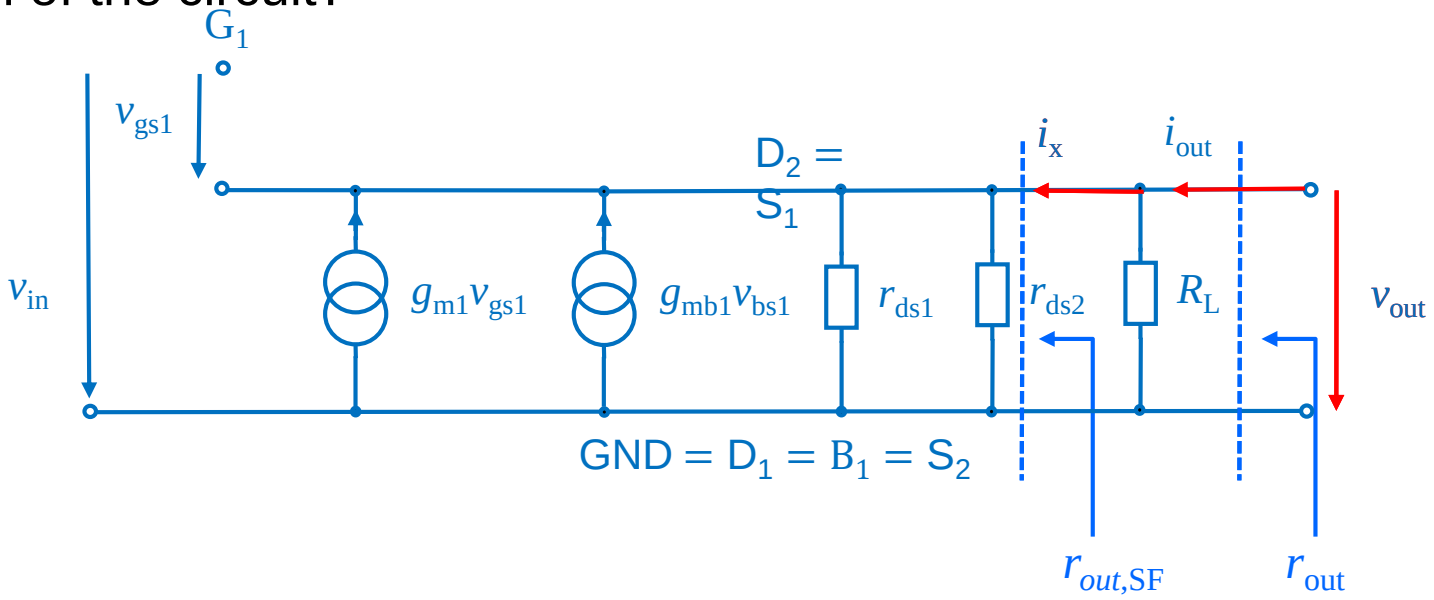
$$v_{\text{gs1}} = -v_{\text{out}}$$

$$v_{\text{bs1}} = -v_{\text{out}}$$

$$r_{\text{out}} = r_{\text{ds1}} \| r_{\text{ds2}} \| \frac{1}{g_{\text{m1}}} \| \frac{1}{g_{\text{mb1}}} \approx \frac{1}{g_{\text{m1}} + g_{\text{mb1}}}$$

Exercise 4 – Task 2

2.6. Repeat task 2.5 while considering the body-effect! Does the body-effect increase or decrease the small-signal gain of the circuit?



$$r_{\text{out,SF}} = r_{\text{ds1}} || r_{\text{ds2}} || \frac{1}{g_{\text{m1}}} || \frac{1}{g_{\text{mb1}}}$$

$$r_{\text{out}} = r_{\text{out,SF}} || R_{\text{L}}$$

$$\Rightarrow \quad r_{\text{out}} = R_{\text{L}} || r_{\text{ds1}} || r_{\text{ds2}} || \frac{1}{g_{\text{m1}}} || \frac{1}{g_{\text{mb1}}} \approx R_{\text{L}} || \frac{1}{g_{\text{m1}}+g_{\text{mb1}}}$$

Exercise 4 – Task 2

2.6. Repeat task 2.5 while considering the body-effect! Does the body-effect increase or decrease the small-signal gain of the circuit?

$$G_{\mathrm{m}} = -g_{\mathrm{m1}}$$

$$r_{\mathrm{out}} \approx R_{\mathrm{L}} \| \frac{1}{g_{\mathrm{m1}}+g_{\mathrm{mb1}}} = \frac{R_{\mathrm{L}}}{(g_{\mathrm{m1}}+g_{\mathrm{mb1}})R_{\mathrm{L}}+1}$$

$$A_{\mathrm{V}} \approx \frac{g_{\mathrm{m1}}R_{\mathrm{L}}}{(g_{\mathrm{m1}}+g_{\mathrm{mb1}})R_{\mathrm{L}}+1}$$

The maximum value of the voltage gain considering the body-effect results in

$$A_{\mathrm{V_{max}}} \approx \lim_{g_{\mathrm{m1}}R_{\mathrm{L}}\to\infty} \left(\frac{g_{\mathrm{m1}}R_{\mathrm{L}}}{(g_{\mathrm{m1}}+g_{\mathrm{mb1}})R_{\mathrm{L}}+1} \right) = \frac{g_{\mathrm{m1}}}{g_{\mathrm{m1}}+g_{\mathrm{mb1}}} < 1$$

The body-effect of the input transistor M_1 decreases the small-signal voltage gain!

Exercise 4 – Task 2

2.7. State the effect of load resistance R_L on the small-signal gain!

$$A_{V_{max}} \approx \lim_{g_{m1}R_L \to 0} \left(\frac{g_{m1}R_L}{(g_{m1}+g_{mb1})R_L + 1} \right) = 0$$

A finite load resistance R_L decreases the small-signal voltage gain of the source-follower.

Exercise 4 – Task 2

2.8. What is the only means to make sure that the load resistance has no impact on the small-signal gain?

$$A_V \approx \frac{g_{m1} R_L}{(g_{m1}+g_{mb1})R_L+1}$$

$$r_{out,SF} \approx \frac{1}{g_{m1}+g_{mb1}}$$

$$\Rightarrow \quad A_V \approx \frac{g_{m1} R_L}{\frac{R_L}{r_{out,SF}}+1}$$

In order to mitigate the effect of a finite load-resistance R_L on the output resistance r_{out}, the following requirement must be met

$$\frac{R_L}{r_{out,SF}} \gg 1$$

Design the output resistance considerably smaller than the load resistance (e.g. $r_{out} < 10R_L$)!

$$(g_{m1} + g_{mb1}) < \frac{1}{10R_L} \quad \Rightarrow \quad A_V \approx g_{m1} r_{out} = \frac{g_{m1}}{g_{m1} + g_{mb1}}$$